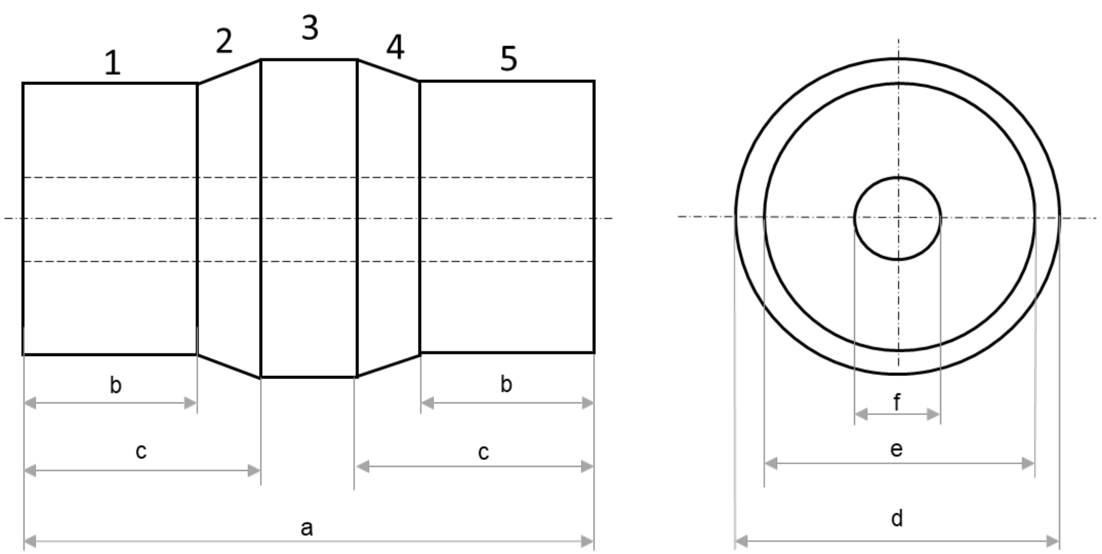
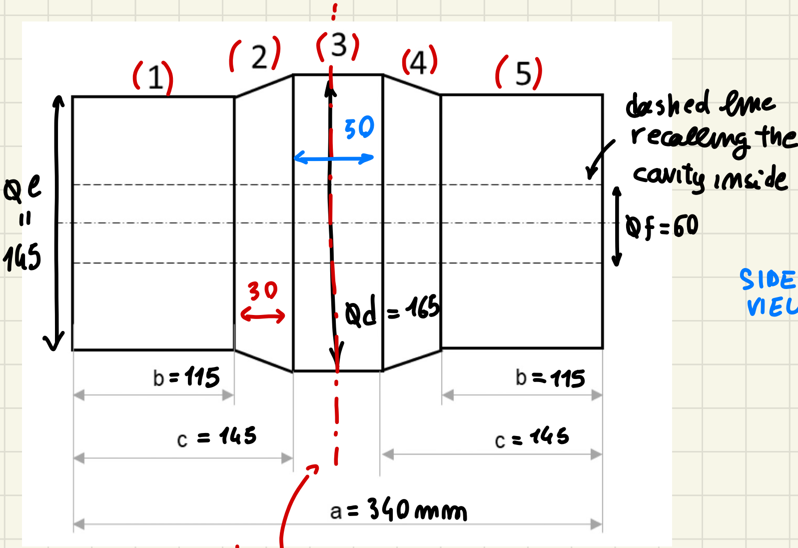
Problem 1 - 06/11/2020

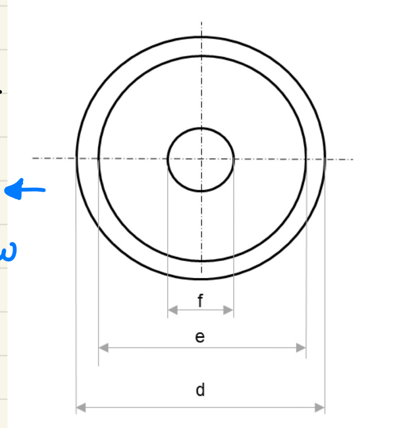
The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of AISI 304 steel (for simplicity, draft angles and fillets/radii are omitted). The casting is divided in 5 elementary geometries (numbered from 1 to 5).

1) Please, study the directional solidification of the metal. Dimensional data: a = 340 mm, b = 115 mm, c = 145 mm, d = 165 mm, e = 145 mm, f = 60 mm.
2) Considering: the casting axis as parting line, $A_s:A_r:A_g$ = 2:2:1, c = 0,6, two gates with an area equal to 700 mm^2 each, a volume of the casting equal to 5 dm^3, a top riser with a volume equal to 3 dm^3, a cope height equal to 300 mm. Design a middle gating system, verifying if the mould filling time is lower than 10 s.





symmetric respect middle plane

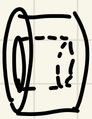


G1 = G5 } same Geometry
G2 = G4 } part

1) DIRECTIONAL solidification

} evaluating MODULUS

• $G_1 = G_5$:



$$\begin{cases} V_1 = \frac{\pi}{4} 145^2 \, 115 - \frac{\pi}{4} 60^2 \, 115 = 1573\,840 \text{ mm}^3 \\ A_1 = \pi 145 \cdot 115 + \pi 60 \cdot 115 + \frac{\pi}{4}(145^2 - 60^2) = 87\,749 \text{ mm}^3 \end{cases}$$

ext + int cylinder + ext face

$\Downarrow$

$$M_1 = V_1 / A_1 = 17.94 = M_5$$ (same of part (1), (5))

[mm]

• $G_2 = G_4$:

$$\begin{cases} V_2 = \frac{1}{12}\pi \left(165^2 + 145^2 + 165 \cdot 145\right) \cdot 30 - \frac{\pi}{4} 60^2 \cdot 30 = 482\,038 \text{ mm}^3 \\ A_2 = \frac{\pi}{2}(145 + 165)\sqrt{30^2 + \left(\frac{165}{2} - \frac{145}{2}\right)^2} + \pi 60 \, 30 = 21054 \text{ mm}^2 \end{cases}$$

external cone + internal cylinder

$$M_2 = V_2 / A_2 = 22.9 \text{ mm} = M_4$$

• G_3 :

a-2c=50

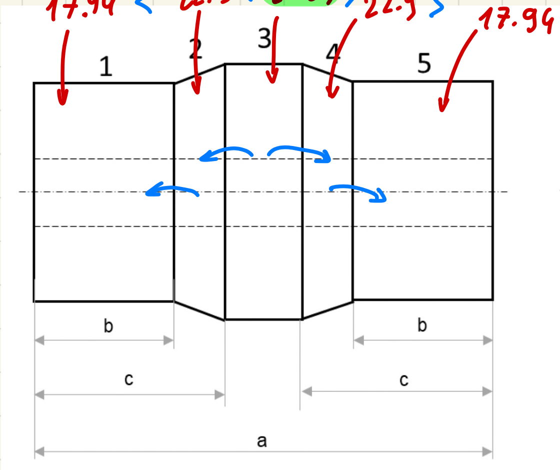
$$\begin{cases} \bar{V}_3 = \frac{\pi}{4}(165^2 - 60^2)\cdot 50 = 927752\ mm^3 \\ A_3 = \pi\,(165+60)\cdot 50 = 35\,343\ mm^2 \end{cases}$$

$$M_3 = \bar{V}_3 / A_3 = 26{,}25\ mm$$

MODULUS

$17.94 < 22.9 < 26.29 > 22.9 > 17.94$

RISER HERE! Where we have bigger modulus...



{ FEASIBLE ? }

evaluating Ratio of M

$$\frac{M_2}{M_1} = \frac{M_4}{M_5} = \frac{22.9}{17.94} = 1.28 \in [1.1 \div 1.3]$$

✓ ON RANGE

$$\frac{M_3}{M_2} = \frac{M_3}{M_4} = \frac{26.25}{22.9} = 1.15 \in [1.1 \div 1.3]$$

✓ on Range safe

solidification starts from

(1),(5) → (2),(4) → (3)

RISER TOP the part 3 → with $M_R > 1.2\, M_3$ at least 20% more!

OUT OF TOPIC

to evaluate the RISER ⇒ $M_R \geq 1.2\ M_3$ (general rule) → better to stay close to $1.2 M_3$

$$M_R = \frac{V_R}{A_R} \geq 1.2\, M_3$$

to compute V_3 ⇝ feed shrinkage @ liquid state

shrink% during solidification? (from pouring to freezing) = s%

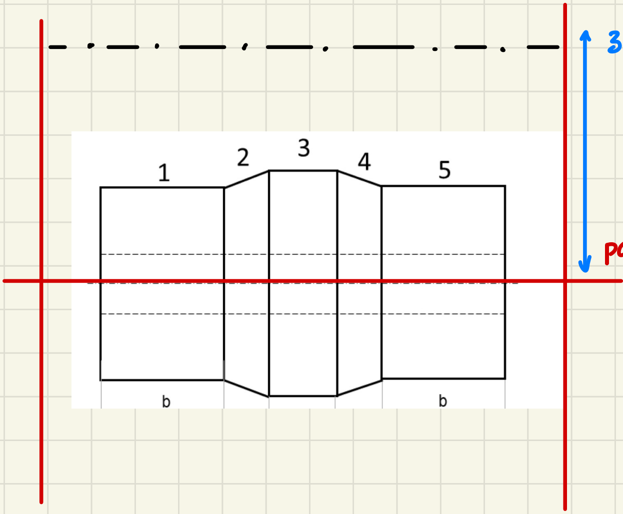
adding 1 RISER → shrinkage of all

$$V_R \geq s\%\ V_{TOT}$$

feed all the object shrink

from that 2 equation you design the RISER!

2) MOLD → design GATING SYSTEM



h_i = 300 mm

middle GATE enters from parting line

parting line

DOWN SPRUE, RISER, GATE

$A_S : A_R : A_G = 2:2:1$

$C = 0.6$

$a_G = 700\ mm^2 \times 2$ gates

$V_{CASTING} = 5\ dm^3$

$V_{RISER} = 3\ dm^3$

$T_{MF} \leq 10s$

design GATING SYSTEM

MIDDLE GATING systems → satisfying that constraint

$$T_{MF} = \frac{V_{TOT}}{A_{BN}\, v} \leq 10s$$

can be computed from DATA

$$V_{TOT} = 5 + 3 = 8\ dm^3$$

A_{BN} smaller area $= A_G = 2 \times 700 = 1400\ mm^2$

$$v = C\sqrt{2g H_{AV}} \rightarrow H_{AV} = \left(\frac{1}{\frac{r_{DRAG}}{\sqrt{h_i}} + \frac{r_{COPE}}{\frac{\sqrt{h_i} + \sqrt{h_f}}{2}}} \right)^2 = \ldots$$

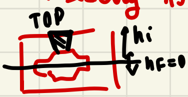
in MIDDLE GATING syst.

$h_i = 300mm$ (total height)

$h_f = 0$ mm

TOP riser height NOT specified

considering h_f as part line



TOP RISER @ hi

$$V_{cope} = \frac{1}{2} V_{CAST} + V_R = \frac{5}{2} + 3 = 5.5\ dm^3$$

$$r_{cope} = V_{cope} / V_{TOT} = 5.5/8 = 0.6875$$

$$V_{DRAG} = \frac{1}{2} V_{CAST} = 5/2 = 2.5\ dm^3$$

$$r_{DRAG} = 2.5/8 = 0.3125$$

... $H_{AV} = \left(\dfrac{1}{\frac{0.3125}{\sqrt{300}} + \frac{0.6875}{\frac{\sqrt{300}+\sqrt{0}}{2}}} \right)^2 = 105.35\,mm$

$\Downarrow$

$v = 0.6\sqrt{2 \cdot 9.81 \frac{105.35}{1000}} = 0.863\,\frac{m}{s}$ $< 1\, m/s$ GOOD VELOCITY ✓

$\Downarrow$

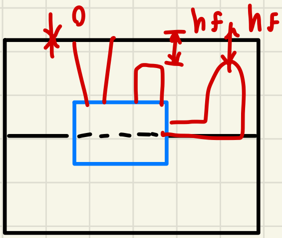
So from HERE

$T_{MF} = \dfrac{8 \times 10^6\,mm^3}{1400\,mm^2} \cdot \dfrac{1}{863\,\frac{mm}{s}} = 6.62\,s$ $< 10\,s$ OK

(Contraint Respected)

NOTE...

changing the RISER



you can have different height

↓ RISER

ASSOCIATION

- TOP ⟶ OPEN
- SIDE ⟶ BLIND

- OPEN
- BLIND ⟶ all inside

CAREFULL !

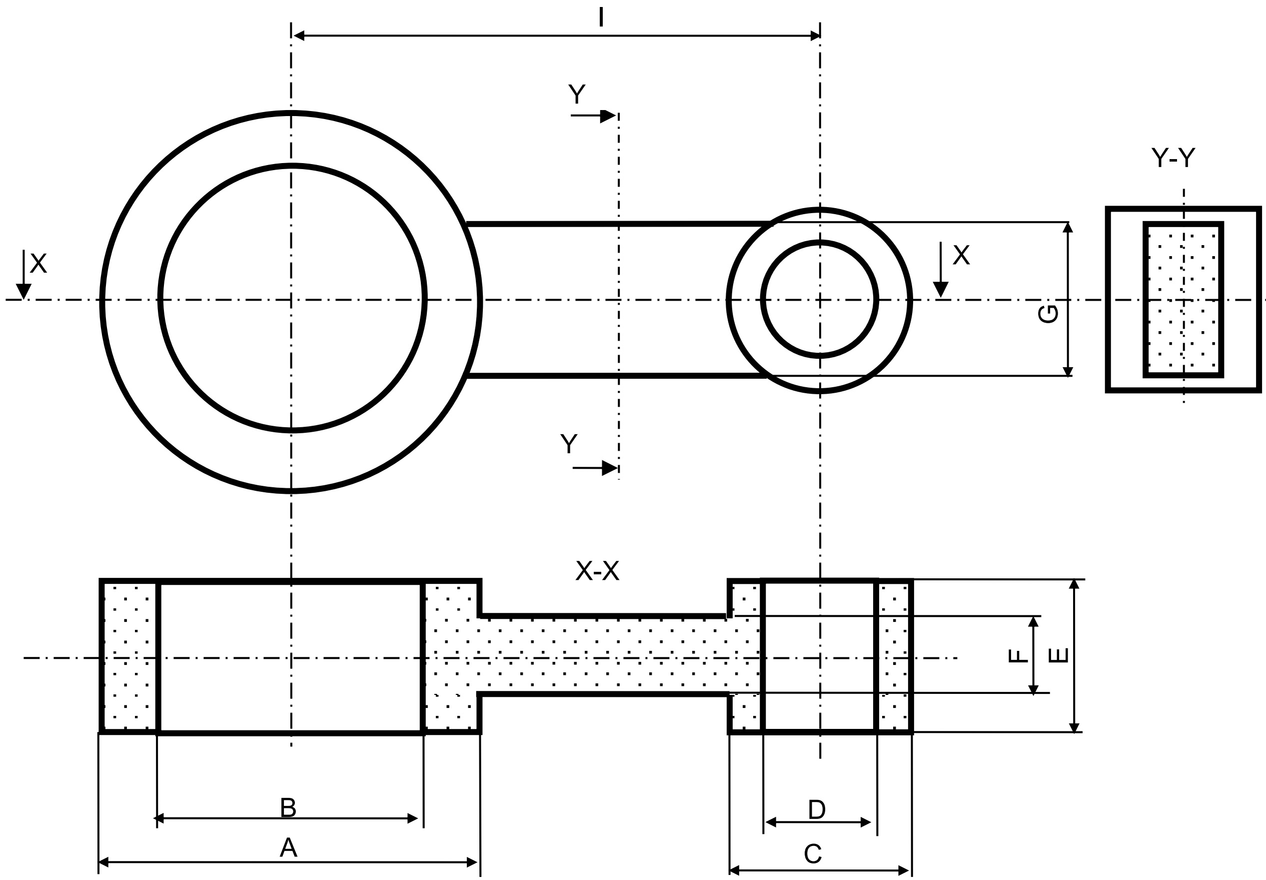
↓

Read well the text

h_F related to the final height reached by the LIQUID..

Problem 2 - 15/01/2021 (1)

The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of steel and will have the following final dimensions after machining: A = 50 mm, B = 35 mm, C = 24 mm, D = 15 mm, E = 20 mm, F = 10 mm, G = 20 mm, I = 70 mm.



Problem 2 - 15/01/2021 (2)

1) Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).
2) Considering the original dimensions (NOT once obtained on (1)), study the directional solidification of the metal. Is the directional solidification feasible? (Consider three elements: Element A is the cylinder of external diameter A; Element C is cylinder of external diameter C, Element B is the middle block to be considered tangent to the cylinders) → simplify Geometry computation!
3) Considering: A_s:A_r:A_g = 4:3:2, c = 0,5, a volume of the casting equal to 36 cm^3, a top riser with a volume equal to 15 cm^3, a distance of the upper plane of the mould from the lower plane of the cavity equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

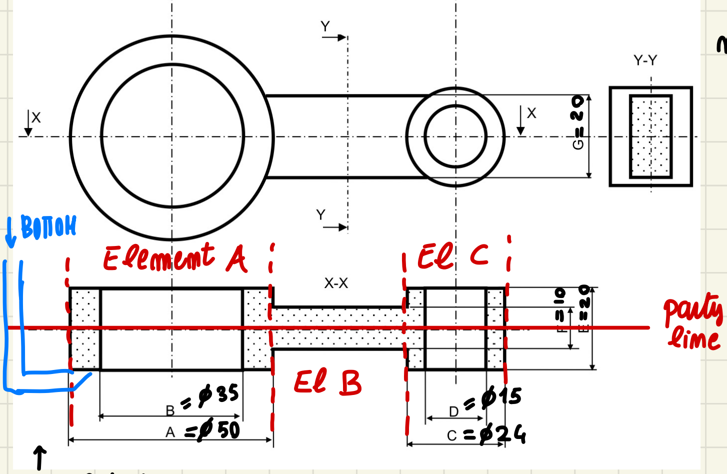
referred to BOTTOM of system at h_i = 200 mm

NOT speaking about PARTING LINE...

If bottom parting line NO! (put on the middle!)

GEOMETRY of CASTING

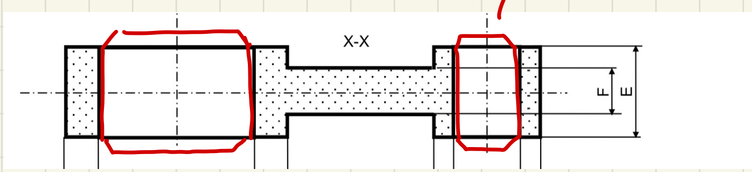
machining allowance
m.a = 2mm
shrinkage allowance
S = 1.4%

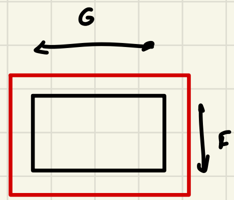


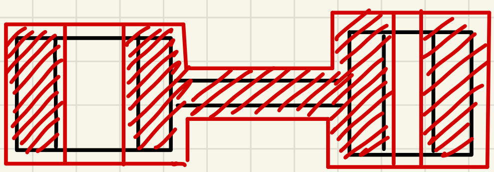
SECTION VIEW

① machining on all surface (all surface need machining)

CORE DESIGN







add material everywhere

change the geometry

to deal with shrinkage ~> machine all surface

$$\Rightarrow A = (50 + 2 \cdot \underbrace{2}_{\text{machining}})(1 + \underbrace{0.014}_{\text{shrinkage allowance}}) = 54 \cdot 1{,}014 = \lceil 54.756 \rceil = 55\,mm$$

as external diameters

external ⌈ ⌉

internal ⌊ ⌋

$$B = (35 - 2 \cdot 2)(1 + 0.014) = \lfloor 31{,}434 \rfloor = 31\,mm$$

$$C = (24 + 2 \cdot 2)(1{,}014) = \lceil 28.392 \rceil = 29\,mm$$

$D = (15 - 2.2)\,1{,}014 = \lfloor 11.154 \rfloor = 11\,mm$

IF TOO small...
avoid to do too small
CORE

Linear dimension

$E = (20 + 2 + 2)\,1{,}014 = \lceil 24{,}336 \rceil = 25\,mm$

$F = (10 + 2 + 2)\,1{,}014 = \lceil 14{,}196 \rceil = 15\,mm$

effect by shrinkage

$G = E = 25\,mm$

I = distance between two axis is NOT effected by machining = $70 \cdot 1{,}014 = 70{,}98 \simeq 71\,mm$

but effect of shrink

we well machine the element ~> TO ROUND IT, on HOLES NOT issue

careful!

holes machined...
if NOT machine → NOT ROUNDING..

ROUND as CLOSEST integer!

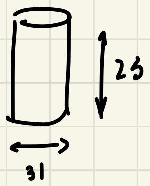
$70{,}98 \simeq 71$
$70{,}12 \simeq 70$
$70{,}70 \simeq 71$
⋮

CORES Design

↓

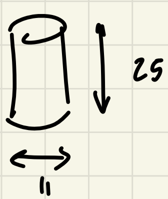
on A → diameter of $B = 31\,mm$ of height $E = 25\,mm$

CORE 1:



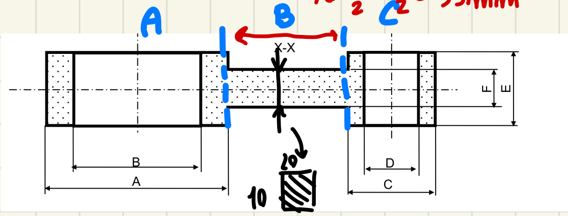
on C of diameter of $D = 11\,mm$ and height $E = 25\,mm$

CORE 2



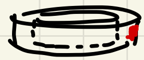
②

$70-\frac{50}{2}-\frac{24}{2}=33mm$



DIRECTIONAL SOLIDIFICATION

Body A) $V_A = \frac{\pi}{4}50^2\cdot 20 - \frac{\pi}{4}35^2\cdot 20 = 20\,028\ mm^3$



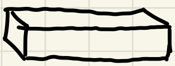
$A_A = 2\cdot\frac{\pi}{4}(50^2-35^2) + \pi\cdot 50\cdot 20 + \pi\, 35\cdot 20 - 10\cdot 20 = 7143\ mm^2$

$\Rightarrow M_A = 2.8\,mm$

Body B)

$V_B = 10\cdot 20\cdot 33 = 6600\ mm^3$

$A_B = 2\cdot 10\cdot 33 + 2\cdot 20\cdot 33 = 1980\,mm^2$ $\Rightarrow M_B = 3.33\,mm$



Body C)

$V_C = \frac{\pi}{4}24^2\cdot 20 - \frac{\pi}{4}\cdot 15^2\cdot 20 = 5\,513\ mm^3$

$\Rightarrow M_C = 1.97\ mm$

$A_C = 2\frac{\pi}{4}(24^2-15^2) + \pi\cdot 24\cdot 20 + \pi\, 15\cdot 20 - 10\cdot 20 = 2802\,mm^2$

So $M_B > M_C$, $M_B > M_A$ $\rightarrow$ last element to solidify is the Body B $\rightarrow$ RISER must be positioned on B

check feasibility:

$M_B/M_C = 3.33/1.97 = 1.69 \notin [1.1, 1.3]$ NOT feasible

$M_B/M_A = 3.33/2.8 = 1.19 \in [1.1, 1.3]$ feasible

shape of object Bad too much difference between B, C

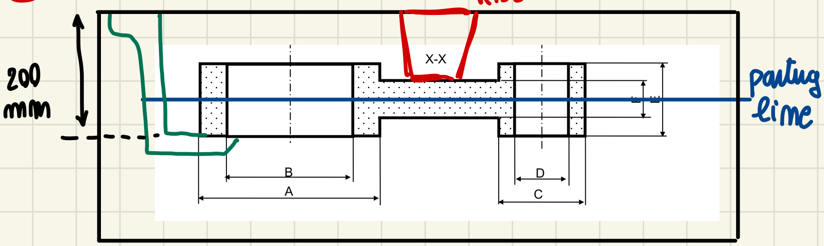
Problem during solid shrinkage (NO PROBLEM during pouring) cooling @ solid state PROBLEM → stresses solid

↓

Deformation or fractures due to stress status

unpredictable behav... ⇒ PROBLEMS!

(3) BOTTOM GATING system DESIGN and verification



$A_s : A_r : A_G = 4 : 3 : 2$ $\quad c = 0.5$

$V_{CASTING} = 36\ cm^3$ $\quad T_{HF} \le 12\ s$

$V_R = 15\ cm^3$

BOTTLENECK → $A_{BN} = A_G$

- define the Area shape (dimensions A_G, A_r, A_s)?

and verify $T_{HF} = \dfrac{V_{TOT}}{A_{BN}\, v} \le 12\ s$

$$v = c\sqrt{2 g H_{av}}$$

$h_i = 200\ mm$

$h_f = 0\ mm$ → TOP OPEN RISER

$$H_{av} = \left(\frac{\sqrt{h_i} + \sqrt{h_f}}{2}\right)^2 = \left(\frac{\sqrt{200} + \sqrt{0}}{2}\right)^2 = 50\ mm$$

$$\Rightarrow v = 0.5\sqrt{2 \cdot 9{,}81 \cdot \frac{50}{1000}} = 0.495\ \frac{m}{s} < 1\ \frac{m}{s}$$ Limited well!

$$A_{BN_{MIN}} = \frac{V_{TOT}}{T_{HF}\, v} = A_G = \frac{36000 + 15000}{12 \cdot 0.495 \cdot 1000} = 8{,}59\ mm^2$$

$[mm]^3$

min value to respect the HF time

$A_r = \frac{3}{2} A_G = 12{,}89\ mm^2$

$A_s = \frac{4}{2} A_G = 17.18\ mm^2$

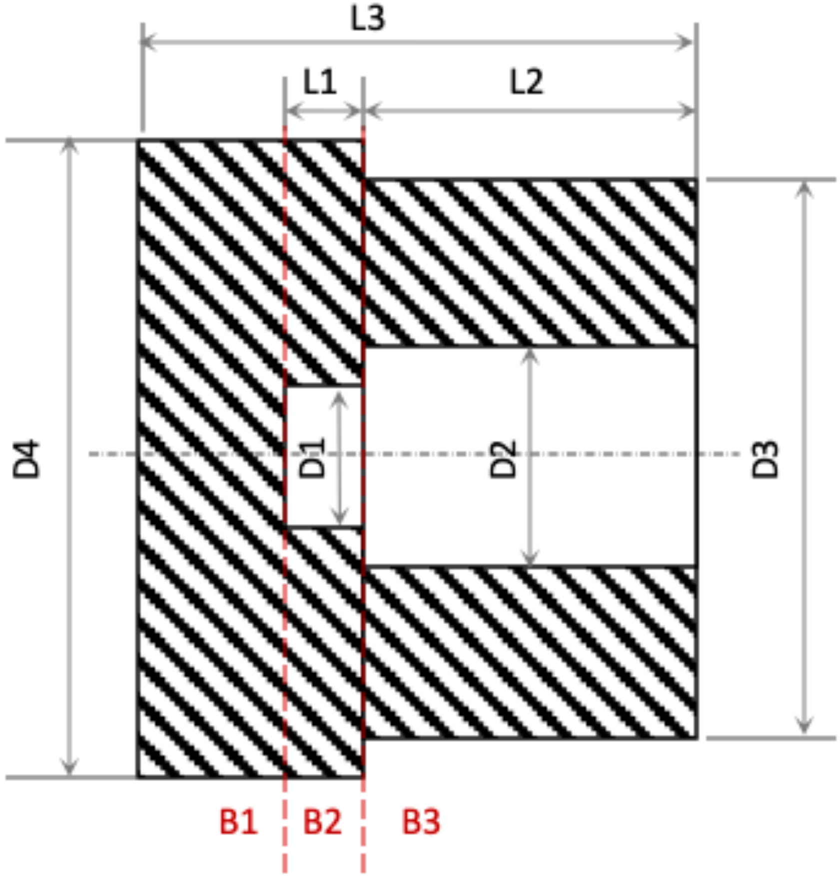
you will choose Number of GATES and area of each gate, runners ecc...

{ Full design of GATING system }

⇒ Real one and reevaluate T_{HF}

Problem 3 - 05/02/2021 (1)

The figure shows an axisymmetric casting to be fabricated by an expendable mould process.
Data: $D1$ = 50 mm, $D1$ = 80 mm, $D3$ = 200 mm, $D4$ = 230 mm, $L1$ = 30 mm, $L2$ = 120 mm, $L3$ = 200 mm.

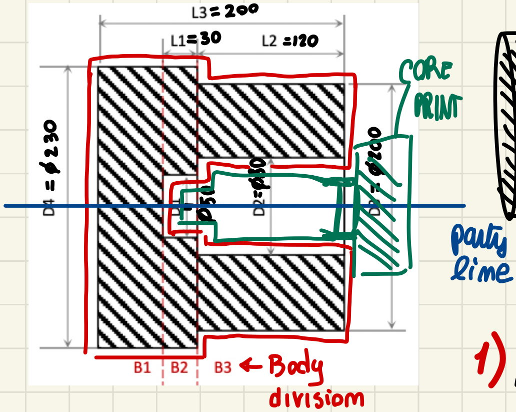


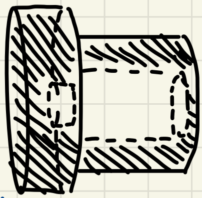
Problem 3 - 05/02/2021 (2)

1) Design the core respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1% (for simplicity, draft angles and fillets/radii are omitted).

2) Considering the original dimensions and the elements as in the figure (B1, B2, and B3), study the directional solidification of the metal. On top of which element should a riser be positioned?

3) Considering: the axis of the casting as parting line, a volume of the casting cavity equal to 6,5 dm^3, a riser with a volume equal to 3 dm^3, a middle gating system with H_{av} equal to 100 mm and c = 0,6. Determine the area of the bottleneck so that the mould filling time is equal to 10 s.





couple of holed cylinders

1) m.a = 2mm

s.a = 1%

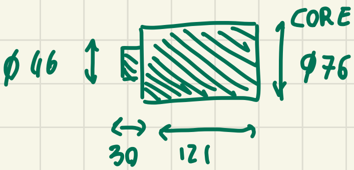
DESIGN THE CORE

$D_1 = (50 - 2\cdot 2)(1 + 0.01) = 46 \cdot 1{,}01 = \lfloor 46{,}46 \rfloor = 46$

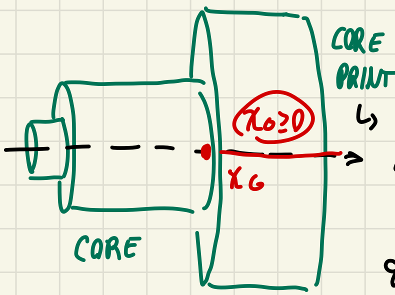
$D_2 = (80 - 2\cdot 2)(1 + 0{,}01) = 76 \cdot 1{,}01 = \lfloor 76{,}76 \rfloor = 76\,mm$

$L_1 = (30 - 2 + 2) \cdot 1{,}01 = 30 \cdot 1{,}01 = \lfloor 30{,}30 \rfloor = 30\,mm$

$L_2 = (120 - 2 + 2) \cdot 1{,}01 = 120 \cdot 1{,}01 = \lfloor 121{,}02 \rfloor = 121\,mm$

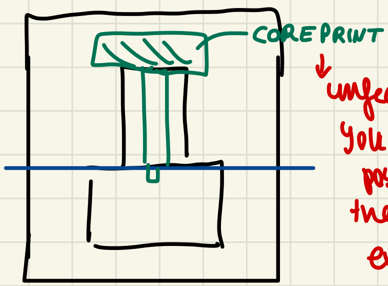


necessary coreprint to let the core be stable



↳ big enough so that the centering gravity is here

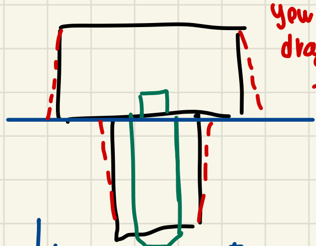
IF I choose to position as



unfeasible! you cannot position the CORE and extract the mold

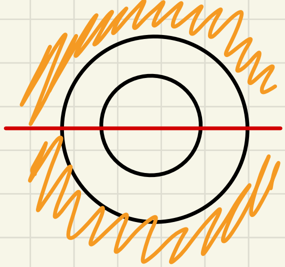
↑ Better core position

Better positioning!



you need draft angles to remove pattern

↳ you'll need to machine the surface

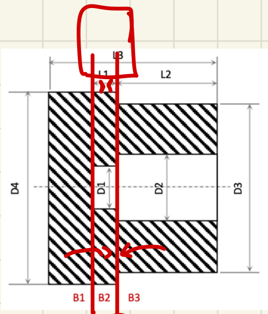


instead with this parting line you don't need draft angles!

pattern separation easy thanks to pattern shape

worst core position BUT better removing

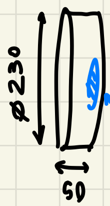
(2)



26.1 (30) 24

last to solidify

Body 1:



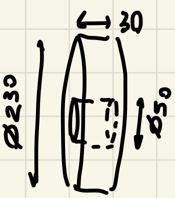
$$V_{B1} = \frac{\pi}{4} 230^2 \cdot 50 = 2077378 \text{ mm}^3$$

$$A_{B1} = \frac{\pi}{4} 230^2 + \underbrace{\pi \cdot 230 \cdot 50}_{\text{ext cylinder}} + \frac{\pi}{4} 50^2 = 79639 \text{ mm}^2$$

also part of heat exchange → through which I exchange heat

$$\Rightarrow M_{B1} = V_{B1}/A_{B1} = 26.1 \text{ mm}$$

Body 2:



$$V_{B2} = \frac{\pi}{4} \cdot 230^2 \cdot 30 - \frac{\pi}{4} 50^2 \cdot 30 = 1187522 \text{ mm}^3$$

$$A_{B2} = \pi \cdot 230 \cdot 30 + \pi \cdot 50 \cdot 30 + \frac{\pi}{4}(230^2 - 200^2) + \frac{\pi}{4}(80^2 - 50^2) =$$

$$= 39584 \text{ mm}^2 \qquad \Rightarrow M_{B2} = 30 \text{ mm}$$

Body 3

$$V_{B3} = \frac{\pi}{4} 200^2 \cdot 120 - \frac{\pi}{4} 80^2 \cdot 120 = 3166725 \text{ mm}^3$$

$$A_{B3} = \frac{\pi}{4}(200^2 - 80^2) + \pi \cdot 200 \cdot 120 + \pi \cdot 80 \cdot 120 = 131947 \text{ mm}^2$$

$$\Rightarrow M_{B3} = 24 \text{ mm}$$

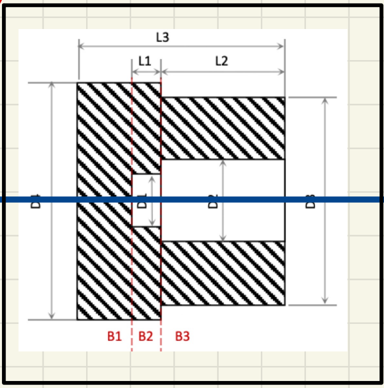
solidification STARTS on (3), then (2) and last one is (2)

$M_{B2}/M_{B1} = 30/26.1 = 1.15 \in [1.1 \div 1.3]$

Feasible

$M_{B2}/M_{B3} = 30/24 = 1.25 \in [1.1 \div 1.3]$

3



$V_{CASTING} = 6.5\ dm^3$

$V_{RISER} = 3 dm^3$

$H_{AV} = 100\ mm \Rightarrow T_{MF} = 10\ s$

$c = 0.6$

$v = c\sqrt{2gH_{AV}} = 0.6\sqrt{2 \cdot 9{,}81 \cdot 100} = 0.84\ \frac{m}{s}$

$< 1\frac{m}{s}$ ok

$V_{TOT} = 6.5+3 = 9.5\ dm^3 =$

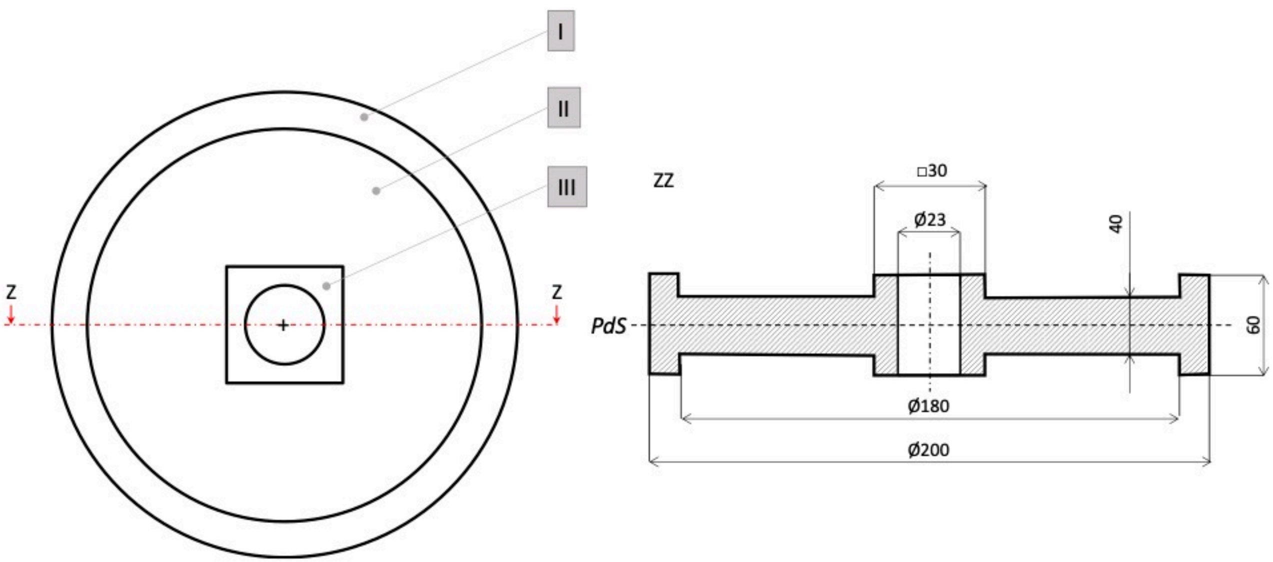
$= 9\,500\,000\ mm^3$

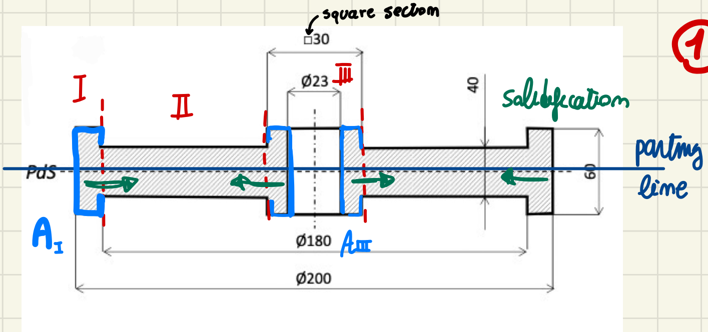
$$A_{BN} = \frac{V_{TOT}}{T_{MF} \cdot v} = \frac{9\,500\,000}{10 \cdot 0{,}84 \cdot 1000} = 1131\ mm^2$$

Problem 4 - 12/07/2021

A steel lift pulley is obtained by sand casting using a middle gating system.
The top view and the cross-section of the main cavity in the mould are shown in the figure, where *PdS* is indicating the parting line.
In the figure the decomposition into elementary geometries is also indicated (draft angles and fillet radii are neglected).

1) Study the directional solidification of the casting.
2) Considering a volume of the entire cavity equal to $1{,}25 \cdot 10^6$ mm^3, a total volume of open risers equal to 400 cm^3, an initial pouring height equal to 100 mm, a friction factor c = 0,6, and a gating system ratio A_s:A_r:A_g = 4:3:2, design the gating system to ensure a mould filling time equal to 10 s.





parting line

① DIRECTIONAL SOLIDIFICATION

Body I:

$$V_I = \frac{\pi}{4}(200^2 - 180^2)60 = 358142\ mm^3$$

$$A_I = 2 \cdot \frac{\pi}{4}(200^2 - 180^2) + \pi \cdot 200 \cdot 60 + \pi \cdot 180(60-40) = 60947\ mm^2$$

$$\Leftarrow \quad M_I = 5.9\ mm$$

Body III: $V_{III} = 30^2 \cdot 60 - \frac{\pi}{4} 23^2 \cdot 60 = 29071\ mm^3$

$$A_{III} = 2 \cdot \left(30^2 - \frac{\pi}{4} 23^2\right) + 4 \cdot 30 \cdot (60-40) + \pi \cdot 23 \cdot 60 = 7704\ mm^2 \Rightarrow M_{III} = 3.8\ mm$$

Body II $V_{II} = \frac{\pi}{4} 180^2 \cdot 40 - 30^2 \cdot 40 = 981876\ mm^3$

$$A_{II} = 2\left(\frac{\pi}{4} 180^2 - 30^2\right) = 49094\ mm^2 \Rightarrow M_{II} = 20\ mm$$

$$M_{II} > M_I > M_{III}$$

last to solidify

FIRST III, then I
cool at end II

DIRECTIONAL solidification → RISERS on part II

$$\frac{M_{II}}{M_I} = \frac{20}{5.9} = 3.4$$

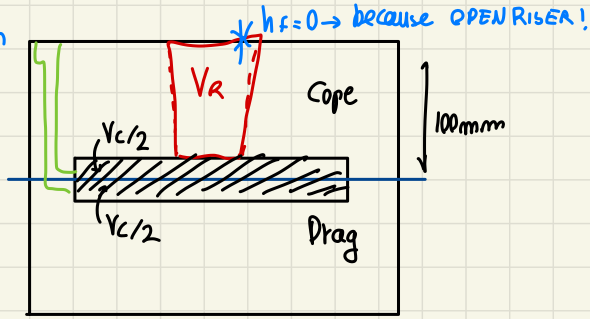
$$\frac{M_{II}}{M_{III}} = \frac{20}{3.8} = 5.26$$

to high! → PROBLEM during solid state cooling

NO

→ NOT FEASIBLE :(

② Middle Gating syst. Design



$$\begin{cases} T_{MF} = 10\,s \\ V_{CAST} = 1.25 \cdot 10^6\ mm^3 \\ V_{R\,(open)} = 400\ cm^3 \\ c = 0.6 \\ A_s : A_R : A_g = 4:3:2 \end{cases}$$

$$V_{TOT} = 1250000\ mm^3 + 400\,000\ mm^3 = 1650000\ mm^3$$

$$r_{Drag} = \frac{1250000/2}{1650000} = 0.38 \qquad r_{cope} = \frac{1250000/2 + 400000}{1650\,000} = 1 - 0.38 = 0.62$$

$\downarrow$

$$H_{AV} = \left(\frac{1}{\frac{r_{drag}}{\sqrt{h_i}} + \frac{r_{cope}}{\frac{\sqrt{h_i} + \sqrt{h_F}}{2}}} \right)^2 = \left(\frac{1}{\frac{0.38}{\sqrt{100}} + \frac{0.62}{\frac{\sqrt{100} + \sqrt{0}}{2}}} \right)^2 = 38.1\,mm$$

$$v = c\sqrt{2gH_{av}} = 0.6\sqrt{2 \cdot 9{,}81 \cdot \frac{38.1}{1000}} = 0.519\ \frac{m}{s} \quad \leq 1\frac{m}{s} \ \text{Good!}$$

1000 ← to have m/s

A_{BN} bottleneck

$\parallel$

$$A_{g\,MIN} = \frac{V_{TOT}}{v\,T_{MF}} = \frac{1650000}{10 \cdot 519\,\frac{mm}{s}} = 318\ mm^2$$

to fully design:

$$A_R = \frac{3}{2} A_g = 477\ mm^2$$

$$A_s = \frac{4}{2} A_g = 636\ mm^2$$

METAL FORMING example (EXAMS)

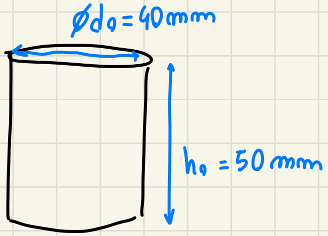
• 06/11/2020 PROBLEM 1

cylindrical sample 6061-O Aluminum alloy

$\sigma = K\varepsilon^n$

$\begin{cases} K = 205\,MPa \\ m = 0.2 \end{cases}$

between two flat dies



1) d_1? if $\varepsilon_1 = -0.16$

2) $h_1 = 43\,mm$
$h_2 = 37\,mm$ 2 steps

$\rightarrow F_1, F_2$? Force each step u, W_I?

3) $u = 0.15\,J/mm^3$

€ energy for 5000 samples?

if €/unit $= 0.1$ €/KWh

① $\varepsilon_1 = -0.16$ COMPRESSION

d_f? $\rightarrow$ considering V = costant

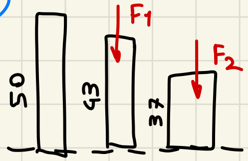
$$V = \frac{\pi}{4} d_0^2 h_0 = \frac{\pi}{4} d_f^2 h_f \qquad d_f = d_0 \sqrt{\frac{h_0}{h_f}}$$

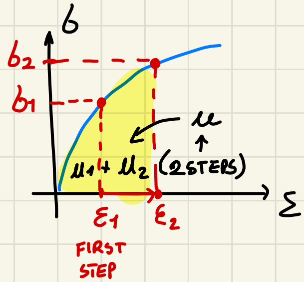
$\Rightarrow$

$$\varepsilon_1 = \ln\left(\frac{h_f}{h_0}\right) \rightarrow h_f = h_0\, e^{-0.16} = 42.61\,mm$$

so $$d_f = 40\sqrt{\frac{50}{42.61}} = 43.33\,mm$$

② 3 STEP Process... $\mu = 0$ F_1, F_2? W_i? (IDEAL)





$\begin{cases} \sigma_1 = K\varepsilon_1^n \longrightarrow F_1 = \sigma_1 A_1 \\ \text{adding deformation } \varepsilon_2 \\ \sigma_2 = K\varepsilon_2^n \longrightarrow F_2 = \sigma_2 A_2 \end{cases}$

area cross section of that situation

by definition:

$\begin{cases} \varepsilon_1 = \ln\left(\frac{43}{50}\right) = -0.15 \\ \varepsilon_2 = \ln\left(\frac{37}{50}\right) = -0.3 \end{cases}$

adding the same deformation of -0.15 for each step... $\Rightarrow$

$\sigma_1 = -205 \cdot 0.15^{0.2} = -140.27\,MPa$

$\sigma_2 = -205\ 0.30^{0.2} = -161.13\,MPa$

so computing the AREA $V_0 = \frac{\pi}{4} 40^2 \cdot 50 = 62832\,mm^3 \longrightarrow$

$A_1 = \frac{V_0}{h_1} = 1461\,mm^2$

$A_2 = \frac{V_0}{h_2} = 1698\,mm^2$

$$\Rightarrow \begin{cases} F_1 = -140.27 \cdot 1461 = -204934\,N = -204.9\,kN \\ F_2 = -161.13 \cdot 1698 = -273599\,N = -273.6\,kN \end{cases}$$

Forces of the 2 steps evaluation

to evaluate the ideal work...

$$W = u_i V_0$$

$$u = \left| \frac{K\,\varepsilon_2^{m+1}}{m+1} \right| = \frac{205 \cdot (0.3)^{0.2+1}}{0.2+1} = 40.3\,MPa = 0.0403\,J/mm^3$$

strain energy density

$$W_i = 0.0403 \cdot 62832 = 2532\,J = 2.5\,kJ$$

3 given

$u = 0.15\ J/mm^3$

$N_s = 5000$ samples

cost $= 0.1$ €/KWh

cost to deform Ns

$$W = V_0 \cdot u = 0.15 \cdot 62832 = 9425\,J$$

$$W_{TOT} = W \cdot 5000 = 47125000\,J$$

$$= 47.125\,MJ \rightsquigarrow W_{TOT} = \frac{47.125}{3.6} = 13.1\,KWh$$

$$\{1\,KWh = 3.6\,MJ\}$$

so $€_{TOT} = W_{TOT}\,\text{cost} = 1.31\,€$

15/01/2021 **PROBLEM 2**

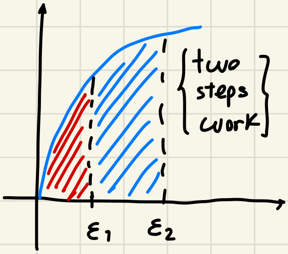
open die forging of cylindrical specimen

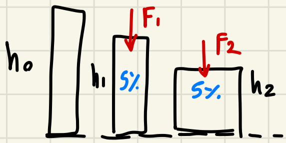
$h_0 = 100$ mm

$d_0 = 50$ mm

$\sigma = K\varepsilon^m$

$\begin{cases} K = 500 \text{ MPa} \\ m = 0.24 \end{cases}$





$h_1 = h_0(1-0.05) = 95$ mm

$h_2 = h_1(1-0.05) = 90.25$ mm

- STRAIN

step 1: (*) $\varepsilon_{0\to1} = \varepsilon_1 = \ln(95/100) = -0.0513$

step 2: (*) $\varepsilon_{1\to2} = \ln(90.25/95) = -0.0513$ (same reduction)

overall: $\varepsilon_{0\to2} = \varepsilon_2 = \ln(90.25/100) = -0.1026 = \varepsilon_{0\to1} + \varepsilon_{1\to2}$

$-0.0513 - 0.0513$

- WORK

step 1: $u_1 = \dfrac{K\varepsilon_1^{m+1}}{m+1} = \dfrac{500\,(0.0513)^{1.24}}{1.24}\,\dfrac{1}{1000} = \dfrac{10.141 \text{ MPa}}{1000} = 0.010141 \dfrac{J}{mm^3}$

$V_0 = \dfrac{\pi}{4} 50^2 \cdot 100 = \text{costant} = 196\,350 \text{ mm}^3$

$W_1 = V u_1 = 0.010141 \cdot 196350 \text{ mm}^3 = 1991 \text{ J}$

step 2: $u_2 = \displaystyle\int_{\varepsilon_1}^{\varepsilon_2} \sigma\, d\varepsilon = \int_{\varepsilon_1}^{\varepsilon_2} K\varepsilon^m d\varepsilon = K\dfrac{(\varepsilon_2^{m+1} - \varepsilon_1^{m+1})}{m+1} = \underbrace{\dfrac{K\varepsilon_2^{m+1}}{m+1}}_{u_{TOT}} - \underbrace{\dfrac{K\varepsilon_1^{m+1}}{m+1}}_{u_1} =$

TOTAL overall — step 1

$= \dfrac{500\cdot(0.1026)^{1.24}}{1.24} - \dfrac{500(0.0513)^{1.24}}{1.24} = \dfrac{23.954 \text{ MPa}}{1000} - \dfrac{10.141 \text{ MPa}}{1000} =$

$= \dfrac{13.813 \text{ MPa}}{1000} = 0.013813 \text{ J/mm}^3$

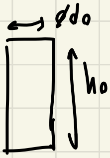
$W_{1\to2} = V u_2 = 0.013813 \cdot 196350 = 2712 \text{ J}$

21/06/2021 PROBLEM 3

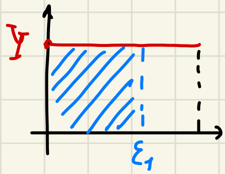
cylindrical speciment

$d_0 = 400\,mm$
$h_0 = 500\,mm$
$Y = 320\,MPa$

perfectly plastic



① $\varepsilon_1 = -0.05$ W_i ?



$u = |\varepsilon_1 \bar{Y}| =$

$= |320 \cdot 0.05| \frac{1}{1000} = \frac{16}{1000} MPa = 0.016 \frac{J}{mm^3}$

$V = \frac{\pi}{4} 400^2 \cdot 500 = 62831853\,mm^3$ $\to W_i = 0.016 \cdot 62831853 = 1005310\,J$

② Neglect Friction h_f ? IF $\varepsilon_1 = -0.06$

$\varepsilon_1 = \ln(h_f/h_0) \to h_f = h_0 e^{\varepsilon_1} = 500 e^{-0.06} = 470.88\,mm$

③ d_{max} ? If $F_{max} = 50\,MN$

because stress fixed $\sigma = Y = 320\,MPa$ as yield stress

$F_{max} = Y A_{max} \to \frac{\pi}{4} d_{max}^2 = \frac{F_{max}}{Y} \to d_{max} = \sqrt{\frac{F_{max}}{Y} \frac{4}{\pi}} = \sqrt{\frac{4 \cdot 50MN}{320MPa\,\pi}} =$

$(F = YA)$

perfectly plastic

$= \sqrt{\frac{4 \cdot 50000000}{320\,\pi}} = 446.031\,mm$

④ $\mu = 0.15$ F ?
$d_1 = 430\,mm$

$A_1 = \frac{\pi}{4} d_1^2 =$

$F_1 = p_{av} A_1 = \underbrace{320 \left(1 + \frac{430\ 0.15}{3 \cdot 432.67}\right)}_{p_{av}} \underbrace{\frac{\pi}{4} 430^2}_{A_1} =$

$= 48779643\,N$

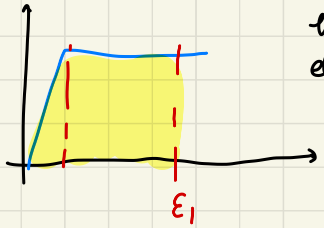
$= 48.8\,MN < 65\,MN$

FEASIBLE!

$p_{av} = Y \left(1 + \frac{\overbrace{2\mu r_1}^{d_1}}{3 h_1}\right)$

$V = \text{costant} = \frac{\pi}{4} d_0^2 h_0 = \frac{\pi}{4} d_1^2 h_1 \to h_1 = \frac{d_0^2}{d_1^2} h_0 = 432{,}67\,mm$

NOTE if a behaviour ELASTIC + PLASTIC



energy on elastic region is negligible